

ALFREDO VARGAS VAZQUEZ

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PROFESSIONAL SUMMARY

Mechatronics Engineering student specializing in Automation with hands-on experience developing technology-driven solutions through software, applied AI, automation, and hardware integration. Experienced in identifying real-world problems, building functional prototypes, deploying user-facing systems, and iterating products based on operational feedback. Strong interest in technical product development, AI-assisted automation, innovation, and scalable technology solutions.

EDUCATION

Universidad Tecnológica de la Sierra Hidalguense

Engineering in Mechatronics — Automation

Hidalgo, Mexico

Aug 2023 – Expected Apr 2027

SELECTED EXPERIENCE & PROJECTS

Gym Access — Biometric Access & Membership Management System

2026

- Developed and deployed a biometric access and membership-management system currently used by a real gym with approximately 600 registered users and an estimated 150 daily check-ins.
- Reduced estimated check-in time from approximately 2–3 minutes to 15–45 seconds by automating user identification and membership-status verification.
- Integrated a DigitalPersona U.are.U 4500 fingerprint reader with a Python/Flask system through a dedicated C#/.NET Framework 4.8 biometric service, solving legacy SDK and runtime compatibility constraints.
- Iterated the product after deployment based on real operational feedback, adding membership management, payments, access history, statistics, and administrative functionality.

Trans-Matic — Quality Inspector Intern / AI Automation Project

Monterrey, Mexico | Apr 2025 – Aug 2025

- Identified a manual quality-reporting workflow that, based on process observations and time measurements, represented an estimated 3,744–5,616 labor-hours annually across the inspection area.
- Designed an AI-assisted automation proof of concept combining voice capture, Whisper speech-to-text, LLM-based data structuring, Google Drive, Zapier, Google Sheets, and Apps Script.
- Analyzed the existing workflow, identified friction points, designed the proposed automation architecture, and evaluated its potential to reduce repetitive administrative work and improve information traceability.

PlantApp — Autonomous Plant Monitoring & Irrigation System

2026

- Developed an autonomous plant-irrigation and monitoring system combining Arduino, sensors, actuators, and a Python/PySide6 desktop application.
- Experimentally calibrated soil-moisture ranges under real conditions and implemented configurable parameters, automatic serial reconnection, and real-time system monitoring.
- Packaged the Python application as a standalone Windows executable while maintaining autonomous Arduino operation when disconnected from the PC.

ENTREPRENEURSHIP & INTERNATIONAL EXPERIENCE

TrepCamp & Santander Universities — Entrepreneurial Experience

ESIC, Madrid, Spain | Jan 17–27, 2026

- Selected as one of approximately 50 participants from nearly 10,000 applicants for an intensive international entrepreneurship and innovation program.
- Received Best Speaker recognition and earned a 100% performance-based scholarship for TrepCamp Simulator in New York City in 2027.

TECHNICAL SKILLS

Programming: Python, JavaScript, C/C++, Java, SQL, HTML/CSS

Applied AI & Automation: Generative AI, Whisper AI, LLM workflows, Zapier, Google Apps Script, APIs, workflow automation

Development: Flask, PySide6, Visual Studio Code, Visual Studio, desktop/web applications

Engineering: Arduino, electronics, MATLAB/Simulink, SolidWorks, sensors & actuators

Data & Productivity: Google Sheets, Google Workspace, Microsoft Excel

LANGUAGES

Spanish: Native | English: B2 — Upper Intermediate | French: B1 — Intermediate